Fayl uzatish protokollari

File Transfer Protocols — FTP / TFTP

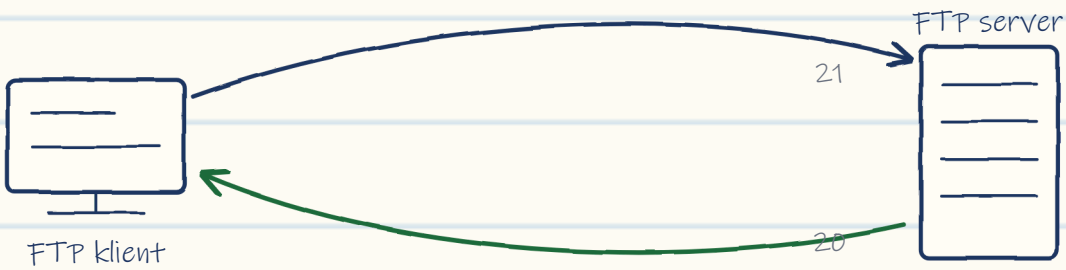
Asosiy atamalar

- FTP — fayl uzatish protokoli, TCP ustida ishlaydi
- Boshqaruv ulanishi — 21-port: buyruqlar va javoblar shu yerdan yuradi
- Ma'lumot ulanishi — 20-port yoki kelishilgan port: faylning o'zi
- Aktiv / passiv rejim — ma'lumot ulanishini server yoki mijoz ochadi
- TFTP — soddalashgan variant: UDP 69, autentifikatsiyasiz
- ASCII / binary — matn yoki o'zgarishsiz (ikkilik) uzatish rejimi

1. FTP asoslari

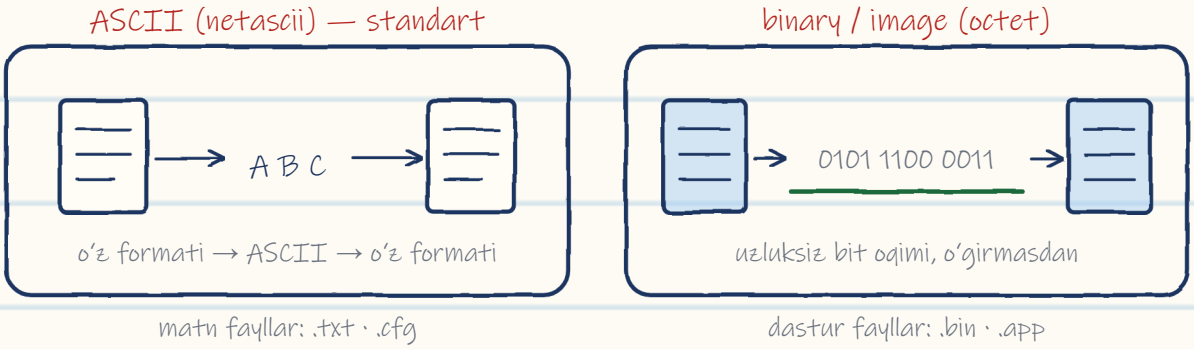
FTP — Internetdagi fayl uzatish standarti. TCP ustida, klient/server.
(File Transfer Protocol)

- ikki TCP ulanish — boshqaruv 21, ma'lumot 20
- login va parol — foydalanuvchi huquqlari beriladi
- 2 fayl rejimi — ASCII · binary
- 2 data rejimi — aktiv (PORT) · passiv (PASV)



1-rasm. Bir seansda ikkita TCP ulanish

2. Fayl uzatish rejimlari

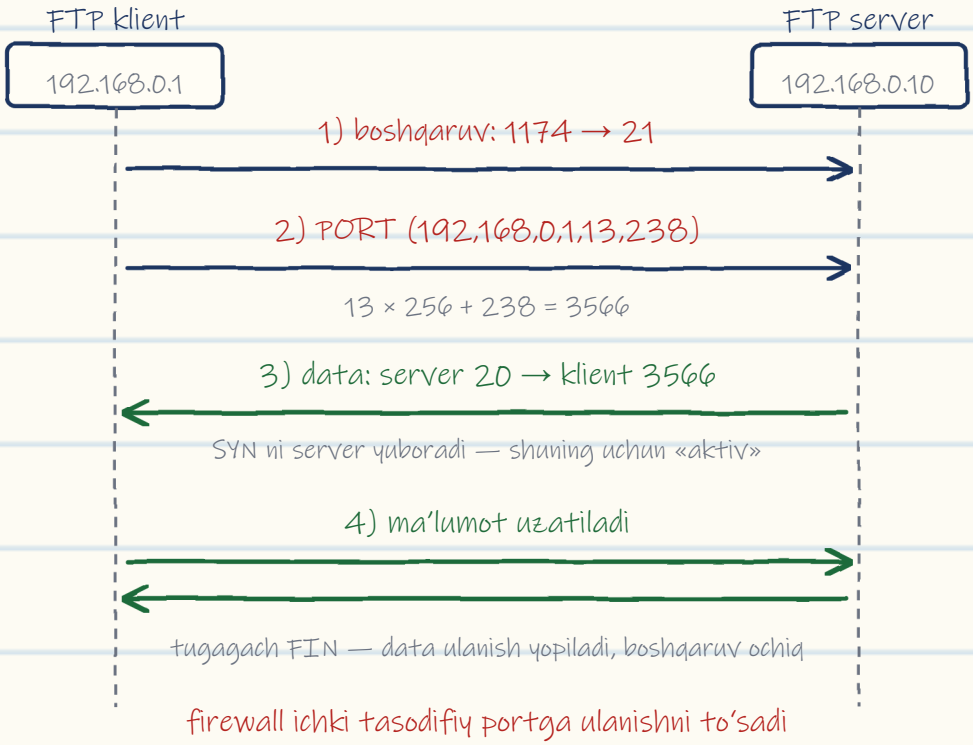


2-rasm. ASCII — matn uchun, binary — dastur uchun

3. Aktiv rejim (PORT)

Klient PORT buyrug'ida o'z vaqtinchalik portini aytadi, data ulanishni server boshlaydi — 20-portdan.

- A1.A2.A3.A4 — klientning IP manzili
- P1, P2 — port: 256 × P1 + P2
- 1174 — boshqaruv porti, 3566 — data porti

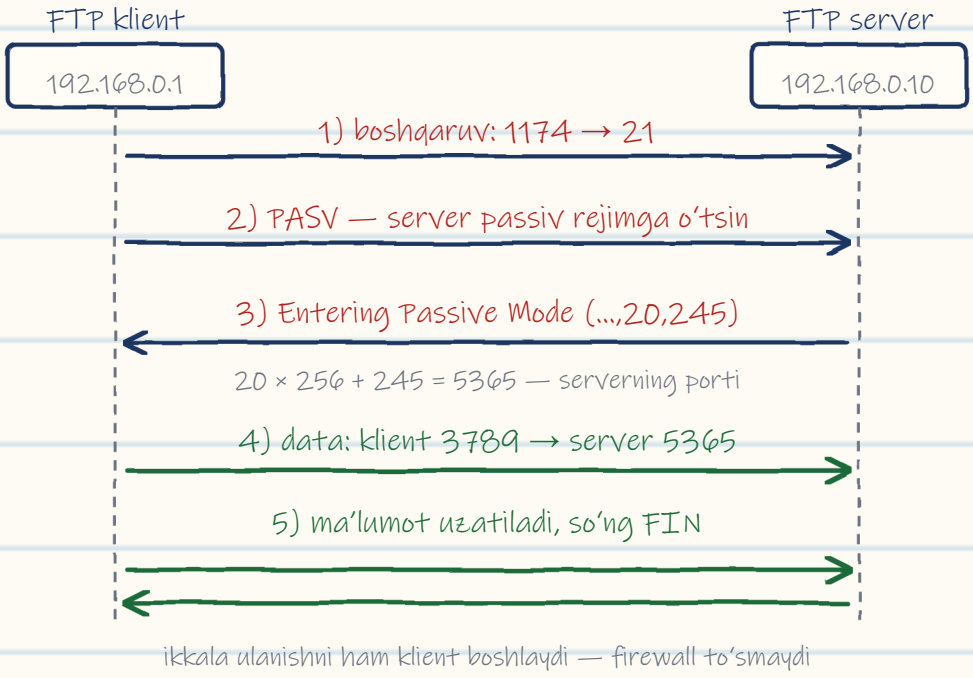


3-rasm. Aktiv (PORT) rejim — to'rt qadam

Data ulanish: fayl yuborish · fayl olish · katalog ro'yxatini yuborish

4. Passiv rejim (PASV)

Klient PASV yuboradi, server javobda o'z vaqtinchalik portini beradi, data ulanishni klient boshlaydi.



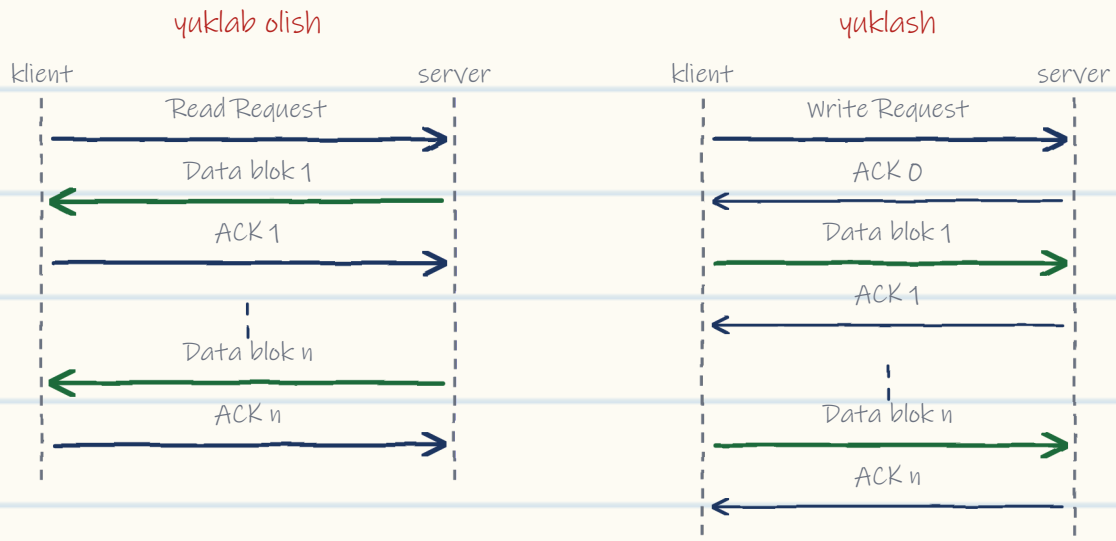
4-rasm. Passiv (PASV) rejim

	Aktiv — PORT	Passiv — PASV
Portni kim aytadi	klient	server
Data ulanishni kim boshlaydi	server (20)	klient
Klient firewall orqasida	muammo	ishlaydi

5. TFTP

TFTP — oddiy fayl uzatish protokoli. (Trivial FTP)

- UDP 69 — bitta port, klient/server
- login yo'q, katalog ro'yxati yo'q, buyruqlar kam
- o'z qayta yuborish mexanizmi — UDP ishonchsiz
- uzatishni faqat klient boshlaydi
- rejimlar: netascii = ASCII, octet = binary



har blok = 512 bayt

oxirgi blok 512 dan kichik → uzatish tugadi

ACK kelmasa — vaqt tugagach qayta yuboriladi

5-rasm. Read Request — olish, Write Request — yuklash

Chegara holat: fayl uzunligi 512 ga bo'linsa, oxirida bo'sh (0 baytli) paket yuboriladi

6. Sozlash



230 User logged in.

Router — FTP server:

- [Router] ftp server enable
- [Router] local-user ftp_manager class manage
- [Router-luser-ftp_manager] password simple 123456
- [Router-luser-ftp_manager] service-type ftp

Router — FTP klient: <Router> ftp *ftp-server*

- ls · get · put · bye · binary · pwd · cd · delete

TFTP klient: <Router> tftp 10.0.0.1 get | put | sget *fayl*

Xulosa

- FTP: TCP · boshqaruv 21 · ma'lumot 20
- Fayl rejimi: ASCII (matn) · binary (dastur)
- Aktiv — data ulanishni server 20-portdan boshlaydi
- Passiv — klient boshlaydi, firewall bilan ishlaydi
- TFTP: UDP 69 · blok 512 bayt · ACK va qayta yuborish

Tekshirish

1. FTP portlari? — 21 boshqaruv, 20 ma'lumot
2. PORT (...,13,238) → port? — 3566
3. Firewall orqasida qaysi rejim? — passiv
4. TFTP transporti va porti? — UDP 69
5. TFTP blok o'lchami? — 512 bayt